SHALINI E

Software Developer | Java | Embedded Systems | IoT

linkedin.com/in/shalinieswaran github.com/Shalini-E

shalinieswaran19042004@gmail.com +91 8148767904

PROFESSIONAL SUMMARY

- ECE graduate with hands-on experience in **Software Development, Embedded Systems, and IoT.**
- Skilled in **Java, Python, C, SQL, HTML, CSS, JavaScript, AWS, STM32, NXP, UART, SPI, ADC, PWM, and sensor interfacing.**
- Strong in **OOP, problem-solving, debugging, teamwork, communication, and adaptability** with a passion for developing reliable software and embedded solutions.

INTERNSHIP

Embedded Engineer-Intern 12/2025 – 05/2026
Visaiyon Energies and Electro Motors Pvt Ltd Coimbatore

- Worked with STM32 and NXP microcontrollers.
- Used UART, SPI, ADC, and sensor interfacing.
- Experience in EV hardware and contactor control.

AI-ML Virtual Internship 07/2024 – 09/2024
EduSkills

- Completed virtual internship in AI & ML.
- Developed machine learning–based solutions.
- Worked on AI-driven business applications.

IN-PLANT TRAINING

Emglitz Technology - AI&IoT 2023

- Learned AI and IoT concepts.
- Worked with IoT modules, sensors, and automation systems.

Durga Tech-Web Development 2022

- Trained in web development fundamentals and front-end technologies.
- Developed mini projects using HTML, CSS, and JavaScript.

PROJECTS

Student Management System using Core Java 2025
Java | OOP | ArrayList | CRUD

- Developed a console-based **Java application** using Object-Oriented Programming concepts.
- Implemented **add, search, update, and delete operations** using ArrayList.
- Applied Java programming and data management concepts to handle student records

Back-EMF-Based Control of BLDC Motors Using the Pulse Width Modulation Technique 2025

Arduino Nano | PWM | Back-EMF | BLDC Motor Control

- Developed a **sensorless BLDC motor control system** using Arduino Nano, PWM, and Back-EMF detection.
- Implemented Back-EMF-based control for **sensorless motor operation** and evaluated motor performance.
- Published the project in **IEEE Xplore** for industrial and IoT applications.

Smart Home Automation by using AI&IoT 2023

IoT | Blynk | Sensors | Automation

- Designed an **IoT-based smart home automation system** using the Blynk platform.
- Enabled remote control and automation of home devices through an IoT-based system.
- Integrated sensors and automation concepts for smart home applications

Form Validation- By using HTML,CSS,Java Script 2022

HTML5 | CSS3 | JavaScript

- Developed a web form with **real-time form validation** using HTML, CSS, and JavaScript.
- Implemented client-side validation to improve **data accuracy and user experience**.
- Applied fundamental **front-end development and JavaScript validation** concepts.

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, C, SQL, JavaScript, HTML, CSS
- **Software Development:** Object-Oriented Programming (OOP), CRUD Operations, Problem Solving, Debugging
- **Embedded Systems:** STM32, NXP Microcontrollers, UART, SPI, ADC, PWM, Sensor Interfacing
- **Version Control:** Git, GitHub
- **Tools & Platforms:** VS Code, STM32CubeIDE, S32 Design Studio, Arduino IDE, Power BI
- **Cloud & Technologies:** AWS (Basic), IoT
- **Database:** SQL

EDUCATION

BE-ECE CGPA: 8.60

K.S.R College of Engineering
Tiruchengode, India
Batch: 2021-2025

CERTIFICATIONS

- Certified Introduction to Amazon Web Services - Edurev 2025
- Certified Java Basics - Edurev 2023
- Certified Advanced Power BI - I-Bacus Tech 2023
- Certified Digi- Trick solving on Automation - Embuzz Technologies pvt Ltd 2022